Gluing through a common point and along trees: complete manuscript costs

Human Compression project

The original is Leemans–Mulpas, *Two gluing methods for string C-group representations of the symmetric groups*, arXiv:2504.17535v1, 24 April 2025 (register L010). All fifteen pages are counted, including references, addresses and the identification banner. We compare them with both complete common papers and the full application. The 23 source records describe one original; the historical numerical sequence and unused preliminaries are not claimed newly recovered.

The complete written comparison. For calibrated selected NFC characters without whitespace, the comparison is

$$R_{\text{written}} = \frac{27081}{12299 + 16117 + 11565} = \frac{27081}{39981} = 0.677347.$$

The replacement written topic is longer in this coordinate. Both common papers are charged once, with every proof, wrapper and reference passage. The earlier Topic B umbrella is not used, and no overlap with another topic is deducted.

Complete document	Pages	Characters
Original L010	15	27081
Common point paper U_1	5	12299
Synchronized tree paper U_2	6	16117
Application A	5	11565
Three diagnostic programs, printed	4	13371

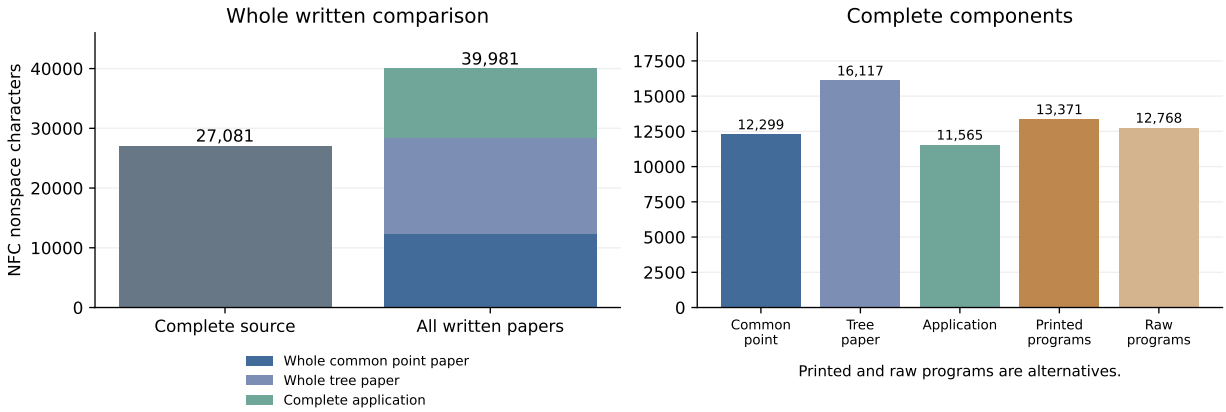
Programs and the conditional application. The full four-page program appendix gives the separate comparison $R_{\text{printed}} = 27081/53352 = 0.507591$. The three complete raw files cost 12768 characters and give $R_{\text{raw}} = 27081/52749 = 0.513394$. These are alternative representations; the printed and raw program costs are never added together. All three programs validate examples and failed weakenings; none is a premise of the uniform written theorems.

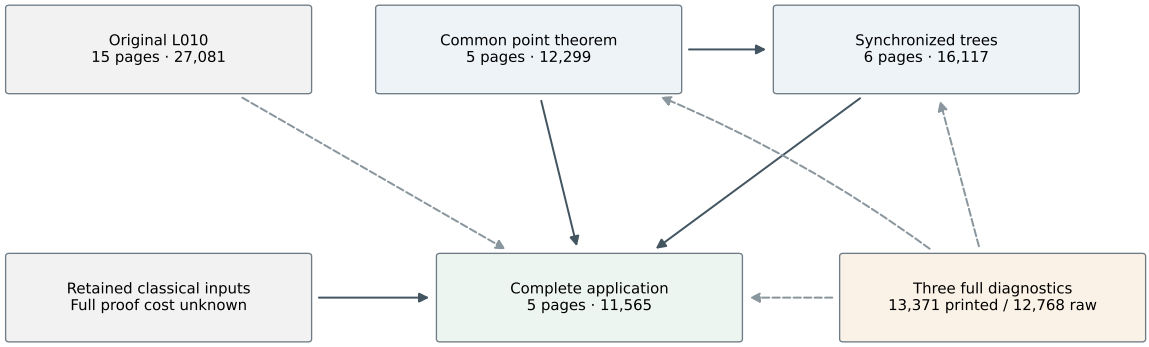
After acquiring *both entire common papers and all three complete declared programs*, the matched application comparison is

$$r \mid (U_1, U_2, V) = \frac{27081}{11565} = 2.341634.$$

This is a conditional acquisition cost, not the whole-topic factor. The three operational drivers cost 13578 characters separately; the two Lean files cost 4214 and contain ten scoped declarations. They do not formalize the complete source or its constructions.

What remains unpriced. Full costs of the retained endpoint sesquiextension theorem and classical endparabolic criterion are unknown. Historical software, complete formalization, and the source diagrams' lines, incidence, colour and layout also remain unknown, not zero. These figures measure available complete texts, not minimum proof complexity or research effort. No new scientific credit is assigned by this numerical packet.





The graph shows complete-document dependencies. Each common paper is paid once even when another paper uses it. Dashed validation and source links are not proof prerequisites. Retained external proof costs stay unknown. There is no node-count ratio or repeated-rise-and-fall claim.

Sensitivity to the unit of length. For calibrated selected NFC text, the whole written, program-inclusive and conditional coordinates are:

Unit	Written	Printed	Raw	Conditional
Whitespace words	0.803	0.713	0.721	2.734
Alphanumeric units	0.828	0.627	0.632	2.841
Nonspace characters	0.677	0.508	0.513	2.342
Collapsed-space characters	0.700	0.540	0.546	2.414
Collapsed-space UTF-8 bytes	0.673	0.531	0.537	2.326

All thirty coordinates are retained in JSON/CSV: five units under NFC and NFKC, for complete raw extraction, furniture-only selection and calibrated selection. Only isolated folios and exact running headers are removed in selected text. Six observed glyph classes produce 175 substitutions; the short stile is a separate encoded component. The curved code-wrap glyph retains its literal comma. The unembedded banner is literal ASCII; unchanged Type0 replacement text has no independent outline-equivalence claim. Full graphical complexity remains outside these text coordinates.